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Assignment 1

Explore following 4 websites and answer the questions

- **bitcoin.org**
- Blockchain.com
- https://coinmarketcap.com/
- https://www.investopedia.com/

One-Word Answerable Questions from bitcoin.org

Bitcoin Basics

1. Bitcoin was introduced by whom? Satoshi Nakamoto
2. Bitcoin transactions are recorded on which data structure? Ledger
3. Bitcoin uses which consensus mechanism? Proof of Work
4. The smallest unit of Bitcoin is called? Satoshi
5. Bitcoin is a **peer-to-peer** type of what system? Network system

Wallets

6. A wallet stores the user's private key.
7. The recovery phrase used in wallets is called a seed phrase.
8. Hardware wallets store keys offline(online/offline).

9. Bitcoin addresses are derived from public keys.
10. What is required to sign a transaction? private key (public/private)

Security

11. Bitcoin prevents double-spending using blockchain.
12. Bitcoin is secured using asymmetric cryptography (asymmetric/symmetric).
13. To run a Bitcoin node, the software required is Bitcoin core.
14. The process of verifying transactions is called mining.
15. The Bitcoin network is described as decentralized (centralized/decentralized).

Mining

16. Miners solve cryptographic puzzles to add blocks.
17. Mining rewards a block with newly created bitcoin.
18. The block time target for Bitcoin is 10 minutes.
19. Mining difficulty adjusts every 2016 blocks.
20. Miners compete to find a valid block hash (hash/signature).

Blockchain Structure

21. Each block contains a merkle root.
22. Blocks are linked using the previous block's hash.
23. A Bitcoin transaction contains inputs and outputs.
24. Unspent transaction outputs are called UTXOS.
25. The first-ever Bitcoin block is called the genesis block.

Network

26. A full node stores the entire blockchain locally.
27. Bitcoin uses which protocol for communication? (TCP/P2P) P2P
28. A light wallet uses SPV mode (SPV/full).
29. Nodes relay information across a P2P network.
30. The maximum Bitcoin supply is 21 million.

Task B1: Explore a Block

Note the following from the **latest block**:

- Block height: 960828
- Block hash: 00000000000000000000f9231726d7648d8a8bef0c0e0e4da85b87d18a331e12
- Timestamp: 03 Aug 2026 05:50:30 UTC
- Number of transactions: 25911
- Miner/pool name: AntPool

Task B2: Explore a Transaction

Open any one Bitcoin transaction.

10. Write the TXID of the transaction. 184089545200cfb6ca35dfe87e7f231dbcf4e2487469485882d9acf56e13aa9b
11. Number of inputs: 5
12. Number of outputs: 4
13. Transaction fee: 4.7 sats/vB
14. How many confirmations does it have? 1

Task B3: Explore an Address

15. Search for any Bitcoin address shown in the block.

16. Record: 1PuJjnF476W3zXfVYmJfGnouzFDAXakkL4

- Total BTC received: 47.159352 BTC
- Total BTC sent: 21.1265 BTC
- Current balance: 8.921578 BTC
- What is the role of a full node? A full node independently downloads, validates, and stores the entire Bitcoin blockchain. It enforces all consensus rules without trusting any third party, and relays valid transactions and blocks to peers, helping maintain the network's decentralisation.

List any two developer resources available on bitcoin.org.

http://developer.bitcoin.org/devguide/block_chain.html

https://developer.bitcoin.org/devguide/mining.html

Explore : https://coinmarketcap.com/

Task C1: Market Information

- Current price of BTC: $63,877.16
- 24-hour % change: +2.13%
- Market capitalization: ~$1.28 Trillion
- Circulating supply: 20,065,615 BTC
- Total/maximum supply: 21,000,000 BTC

Task C2: Historical Data

- What was the price of Bitcoin one year ago? ~$117,815
- Note the highest and lowest prices in the last 30 days. ~$71,360 , ~$60,074

Explore https://www.investopedia.com/

1. Why is Bitcoin considered volatile?

Bitcoin is driven by speculation and sentiment rather than underlying cash flows. Its relatively small market size means large holders can swing prices significantly. Regulatory news, 24/7 trading, and the lack of any intrinsic value floor all amplify price swings in both directions.

2. List two investment risks mentioned.
    a. Regulatory risk — governments can suddenly ban or restrict Bitcoin trading
    b. Security risk — exchanges can be hacked or go bankrupt, and lost private keys mean permanent loss of funds

🔹 Part E: Reflection & Analysis

1. Which website gave you the most clear understanding? Why?
bitcoin.org was the easiest to understand. It doesn't try to sell you anything or overwhelm you with price charts — it just explains what Bitcoin is and how it works in plain language.

2. In one paragraph, explain how blockchain ensures trust without a central authority.
Instead of relying on a bank or government to keep records, blockchain lets thousands of computers around the world maintain the same copy of every transaction. Each block is locked to the previous one using a unique code (hash), so tampering with even one record would break the entire chain. Since no single person or organization controls it, there's no way to cheat the system — the network itself acts as the authority.